# IS1212 : Probability and Statistics

*K S Saubhagya*
*Lecturer (Probationary)*
*Department of Statistics, Faculty of Science, University of Colombo*
*saubhagya@stat.cmb.ac.lk*

# Poisson Distribution

A discrete r.v. X having pdf of the form,

$$P(X=x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$ for x=0,1,2,3….to infinity

Where λ can take any positive value, is said to follow the Poisson distribution.

# Expectation and Variance

- If $X \sim Po(\lambda)$ then it can be shown that $E(X) = \lambda$ and $Var(X) = \lambda$.
- Uses of the Poisson Distribution.
- If an event is randomly scattered in time/space and has mean number of occurrence $\lambda$ in a given time/space and if X is the r.v. 'the number of occurrences in the given interval, then $X \sim Po(\lambda)$.

# Examples:

- Car accidents on a particular stretch of a road in one day
- Accidents in a factory in one week
- Telephone calls made to a switch board in a given minute
- Insurance claims made to a company in a given time

# Examples

(1) If X~ Po(2), find P($X = 4$) and P($X \geq 3$)

# Solution

(1) If X~ Po(2), find P($X = 4$) and P($X \geq 3$)

$$P(X=4) = \frac{e^{-2}\,2^4}{4\,!} = 0.0902$$

# Solution

(1) If X~ Po(2), find P($X = 4$) and P($X \geq 3$)

$$P(X>=3) = 1 - [P(X=0)+P(X=1)+P(X=2)]$$

$$= 1 - \left[ \frac{e^{-2}\,2^0}{0!} + \frac{e^{-2}\,2^1}{1!} + \frac{e^{-2}\,2^2}{2!} \right]$$

$$= 0.323$$

# Examples

(1) The mean number of bacteria per milliliter of a liquid is known to be 4. Assuming that the number of bacteria follows a Poisson distribution, find the probability that, in 1ml of liquid, there will be

(a) No bacteria

(b) 4 bacteria

(c) less than 3 bacteria

# Solution

X~ Po(4),

(a)No bacteria

$$P(X=0) = \frac{e^{-4}\,4^{0}}{0\,!} = 0.0183$$

(b) 4 bacteria

$$P(X=4) = \frac{e^{-4}\,4^{4}}{4\,!} = 0.195$$

(c) less than 3 bacteria

# Solution

$X \sim Po(4)$,

(c) less than 3 bacteria

$P(X<3) = P(X=0)+P(X=1)+P(X=2)$

$$= \frac{e^{-4}4^0}{0!} + \frac{e^{-4}4^1}{1!} + \frac{e^{-4}4^2}{2!} = 0.238$$

# Example

(2) Using the data of previous exercise find the probability that in 3ml of liquid there will be less than 2 bacteria.

# Solution

(2) Using the data of previous exercise find the probability that in 3ml of liquid there will be less than 2 bacteria.

Let Y be the number of bacteria in 3ml of liquid Y~Po(12)

$$Y \sim Po(12)$$

$$P(Y<2) = P(Y=0)+P(Y=1)$$

$$= \frac{e^{-12}\, 2^0}{0\,!} + \frac{e^{-12}\, 12^1}{1!} = 7.99 \times 10^{-5}$$

# Geometric Distribution

Consider a sequence of iid Bernoulli trials with P({S})=p

Variable of interest: Number of trails that has to be performed until the first success

- Random variable: X is the number of trials on which the first success occurs
- Distribution: For X=x there must be a run of (x-1) failures followed by a success.
- So,

$$P(X=x)= p(1-p)^{x-1},\ x=1,2,3\ldots\ldots;\ 0<p<1$$

# Moments of Geometric Distribution

$$\text{Mean} = \mu = \frac{1}{p}$$

$$\text{Variance} = \sigma^2 = \frac{(1-p)}{p^2}$$

# Example

Suppose that the probability of having a male or a female child is equal.,

Find the probability that family's fourth child is their first son.

$$P= \frac{1}{2}^{3} * \frac{1}{2} = 0.0625$$

# Negative Binomial Distribution

This is a generalisation of the geometric distribution

- Random variable: X is the number of trials on which the $k^{th}$ success occurs where k is a positive integer
- Distribution: P(X=x)= $\binom{x-1}{k-1} p^k (1-p)^{x-k}$, x=k, k+1, …..,0<p<1

We say that X has negative binomial (k,p) distribution.

The probabilities satisfy the recurrence relationship

$$P(X=x)=\binom{x-1}{x-k}(1-p)P(X=x-1)$$

Geometric distribution is special case of this random variable when k=1

# Moments of Negative Binomial Distribution

$$\text{Mean} = \mu = \frac{k}{p}$$

$$\text{Variance} = \sigma^2 = \frac{k(1-p)}{p^2}$$

# Example

An oil company conducts a geological study that indicates that an exploratory oil well should have a 20% chance of striking oil. What is the probability that the third strike comes on the seventh well drilled?

# Solution

X=7, k=3, p=0.2

$$P(X=x)=\binom{x-1}{k-1}p^k(1-p)^{x-k}$$

$$P(X=7)=\binom{7-1}{3-1}0.2^3(1-0.8)^{7-3}= 0.049$$

# Hypergeometric Distribution

- This is the finite population equivalent of the binomial distribution, in the following sense.
- Suppose objects are selected at random, one after the another, without replacement, from a finite population consisting of k successes and N-k failures. The trials are not independent.
- Random variable: X is the number of 'successes' in a sample size n from a population of size N that has k 'successes' and (N-k) failures.

Distribution:

$$P(X=x)=\frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}},\ x=1,2,3,\ldots\ldots,0<p<1$$

# Moments of Hypergeometric Distribution

Mean = $\mu = \frac{nk}{N}$

Variance = $\sigma^2 = \frac{nk(N-k)(N-n)}{p^2}$

# Example

A deck of cards contains 20 cards: 6 red cards and 14 black cards. 5 cards are drawn randomly without replacement. What is the probability that exactly 4 red cards are drawn?

# Solution

The probability of choosing exactly 4 red cards is :

$$P(4 \text{ red cards}) = \frac{\#\text{ samples with 4 red cards and 1 black card}}{\#\text{ of possible 4 card samples}}$$

$$P(X=4) = \frac{\binom{6}{4}\binom{20-6}{5-4}}{\binom{20}{5}} = 0.0135$$

Any questions?